

TPS63070 3.3V Buck-Boost Module

Project Reference — 2 V to 16 V Input → Stable 3.3 V Output

1. Overview

This module is a small breakout board built around Texas Instruments' **TPS63070** buck-boost converter IC. (The Amazon listing's "XL63070" name is a marketing label — the actual silicon is the TI TPS63070.) Unlike a plain buck (step-down) or boost (step-up) regulator, a buck-boost converter automatically chooses the right mode based on whether the input is above or below the desired output, giving you a stable rail across a wide input range.

On this board, the output is fixed at **3.3 V** with input accepting anywhere from 2 V to 16 V. That makes it ideal for running 3.3 V microcontrollers and sensors from sources whose voltage drifts above and below 3.3 V over time — most notably a single-cell lithium-ion battery (roughly 3.0–4.2 V), where a plain LDO or boost converter alone would not suffice.

Key capabilities:

- Wide input voltage range: 2 V to 16 V (single Li-ion, 2× AA, 9 V battery, 12 V supply)
- Fixed 3.3 V output, regulated whether input is above or below 3.3 V
- Up to ~2 A continuous output current in both buck and boost modes
- Up to ~95 % efficiency at typical operating points
- Automatic Power Save (PFM) mode at light loads for low quiescent current
- Forced PWM mode available for low-noise / low-ripple applications
- Synchronous rectification (no external Schottky diode needed)
- Internal soft-start, current limit, thermal shutdown, and short-circuit protection
- Fixed switching frequency of 2.4 MHz (PWM mode) — small external components

2. Specifications

Parameter	Value
Controller IC	Texas Instruments TPS63070
Input voltage range	2.0 V to 16.0 V
Output voltage (this module)	3.3 V (fixed)
Output current (max, continuous)	≈ 2 A (boost or buck mode)
Switch peak current (chip rating)	3.6 A typical
Switching frequency	2.4 MHz (PWM mode, fixed)
Efficiency (typical)	Up to ~ 95 % @ 3.3 V output
DC accuracy	± 1 % (PWM) / $+3$ % / -1 % (PFM)
Operating modes	Auto PWM/PFM (board default)
Operating temperature (chip)	-40 °C to $+125$ °C (junction)
Module input/output pads	VIN, GND $\times 2$, VOUT (through-hole)
Output ripple	Low (small ceramic L-C output filter)
Protection features	Soft-start, OCP, thermal shutdown, short-circuit

Warning: never exceed **16 V on VIN** — the TPS63070's absolute maximum is 17 V on the input pin and exceeding it can permanently damage the IC. Also avoid reverse polarity: most low-cost modules don't include reverse-polarity protection, and a backwards battery will instantly destroy the chip.

3. Board Pinout

The module exposes four labeled through-hole pads — two on each end. There are no signal pins (EN, FB, PG, etc.) brought out on this version: the chip's enable is tied to VIN internally, the feedback divider is set on the board for fixed 3.3 V, and the converter starts as soon as input voltage is applied.

Pad	Side	Function
VIN	Right	Input voltage, 2 V to 16 V. Connect your battery / wall supply / source here.
GND	Right	Ground (input return). The two GND pads are connected together on the board.
VOUT	Left	Regulated 3.3 V output. Up to ~2 A continuous to your load.
GND	Left	Ground (output return). Use this GND pad for your load to keep return paths short.

Tip: always connect both your input source's GND *and* your load's GND — they are joined on the board, but using the GND pad on the same side as your VIN/VOUT pad keeps current loops small and reduces noise. For high-current loads (close to 2 A), use thick wire and short leads.

Board Layout Reference

Top-down view of the module (component side facing up). Output pads on the left, input pads on the right.



Component side view. The two GND pads are connected internally — use whichever is closer to your wiring.

4. Typical Wiring & Use Cases

The buck-boost converter excels exactly where simpler regulators fail — when your source voltage **crosses** the desired output voltage during operation. Below are the most common scenarios this module is bought for, and how to wire each one.

4.1 Single-cell lithium battery → 3.3 V MCU/sensor

A Li-ion or LiPo cell ranges from ~4.2 V (full) down to ~3.0 V (empty). A plain LDO would drop out at the low end; a boost-only converter can't handle the high end. The TPS63070 transitions seamlessly between buck and boost as the cell discharges.

Connection	Wire to
Battery (+)	VIN
Battery (–)	GND (input side)
MCU 3V3 rail / VCC	VOUT
MCU GND	GND (output side)

4.2 2× AA or 2× AAA → 3.3 V

Two alkaline cells fresh = ~3.2 V, near end of life = ~1.8 V. The module will boost up to 3.3 V across the full discharge curve, drawing more input current as the batteries deplete.

Connection	Wire to
Battery pack (+)	VIN
Battery pack (–)	GND (input side)
Load VCC	VOUT
Load GND	GND (output side)

4.3 9 V or 12 V supply → 3.3 V (buck mode)

When input is well above output, the chip operates as a synchronous buck. Compared to an LDO (say, an AMS1117), efficiency is dramatically higher (~90 % vs. ~28 % at 12 V → 3.3 V) — meaning much less wasted heat at the same load current.

Connection	Wire to
12 V supply (+)	VIN
12 V supply (–)	GND (input side)
Load VCC	VOUT
Load GND	GND (output side)

5. When to Use This vs. Other Regulators

This module solves a very specific problem. Picking the right regulator for the job avoids wasted cost, wasted board space, and unnecessary noise.

Scenario	Best choice	Why
Wall supply 5 V $\rightarrow$ 3.3 V, low current (<300 mA)	LDO (AMS1117-3.3, MCP1700)	Cheap, quiet, no inductors. Heat is acceptable at low current.
Wall supply 12 V $\rightarrow$ 3.3 V at >300 mA	Buck converter (e.g. MP1584, LM2596)	LDO would burn $\sim$ 3 W as heat. Plain buck is enough since input is always > output.
Boost USB 5 V to 12 V	Boost converter (e.g. MT3608)	Input is always below output, so a simple boost works.
Single Li-ion battery $\rightarrow$ 3.3 V	This module (TPS63070 buck-boost)	Battery crosses 3.3 V during discharge (4.2 V $\rightarrow$ 3.0 V). Only buck-boost can handle it.
2 $\times$ AA $\rightarrow$ 3.3 V	This module	Source starts above and ends below 3.3 V.
Solar panel / variable source	This module	Input voltage swings widely; buck-boost keeps output rail stable.
Need clean low-noise rail for ADC / RF	LDO downstream of this module	Switching converters have ripple. Cascade an LDO to scrub it.

6. Operating Modes (PWM vs PFM)

The TPS63070 automatically switches between two modes based on load current to keep efficiency high across a wide range:

- **PWM (Pulse-Width Modulation)** — at moderate to heavy loads, the converter runs at a fixed 2.4 MHz switching frequency. Output ripple is predictable and confined to that frequency, which is easy to filter. DC accuracy is $\pm 1\%$.
- **PFM (Power Save / Pulse-Frequency Modulation)** — at very light loads, the chip skips switching cycles to save energy. Quiescent current drops dramatically, but switching events become irregular and can produce audible noise from the inductor in extreme cases. DC accuracy widens to $+3\%$ / -1% .
- **Buck-boost transition** — when input is near 3.3 V, the controller smoothly blends between buck-mode switching and boost-mode switching using its average-current control loop. There is no dropout zone: the output stays at 3.3 V even when VIN momentarily equals VOUT.

The Amazon listing mentions "adjustable PWM and PFM modes" — on the bare TPS63070 chip, PFM/PWM selection is controlled by the PS/SYNC pin. On this board, that pin appears to be tied for automatic mode (the most common configuration). If you specifically need forced-PWM operation (lowest noise, constant frequency) and the board doesn't expose a control pad, you would need to use the bare chip or a different breakout.

7. Efficiency & Power Budget

Real-world efficiency depends on input voltage, output current, and operating mode. The numbers below are typical and based on TI's datasheet curves for the TPS63070 with a 3.3 V output. Use them to estimate input current and battery life.

Input voltage	Mode	Output current	Approx. efficiency
3.6 V (Li-ion full)	Boost (slight)	500 mA	~93 %
3.6 V (Li-ion full)	Boost (slight)	1 A	~91 %
3.0 V (Li-ion empty)	Boost	500 mA	~90 %
3.0 V (Li-ion empty)	Boost	1 A	~88 %
5.0 V (USB)	Buck	500 mA	~93 %
5.0 V (USB)	Buck	1 A	~92 %
12 V	Buck	500 mA	~88 %
12 V	Buck	1 A	~85 %

Quick formula

To estimate the input current I_{in} from a given output current I_{out} (and chosen efficiency η):

$$I_{in} = (V_{out} \times I_{out}) / (V_{in} \times \eta)$$

Example: 3.7 V Li-ion driving an MCU + sensors at 200 mA from VOUT, assuming 92% efficiency:

$$\begin{aligned} I_{in} &= (3.3 \text{ V} \times 0.20 \text{ A}) / (3.7 \text{ V} \times 0.92) \\ &= 0.66 \text{ W} / 3.404 \text{ W per amp} \\ &\approx 0.194 \text{ A} \rightarrow \text{about 194 mA from the battery} \end{aligned}$$

Worked the other way, a 2000 mAh Li-ion would run that load for roughly $2000 / 194 \approx 10.3$ hours (ignoring discharge curve effects).

8. Design Notes & Gotchas

Capacitor placement

Add a low-ESR ceramic capacitor (10–22 μF , X5R or X7R) at the load if your load is more than a few centimeters of wire away from the module. Long leads add inductance that the chip's output filter wasn't designed to compensate for — symptoms include oscillation, ringing, or instability under load steps.

Input source impedance

For inputs delivered through long thin wires (test leads, breadboard jumpers), add a 47–100 μF electrolytic at VIN. The buck-boost can pull large pulses of current, and a high-impedance source will sag enough to trigger the chip's UVLO and reset.

Inrush current

At startup, the chip soft-starts and limits the input current to about 1 A while charging the output capacitor. After power-good, the limit returns to its full ~ 3.6 A switch peak. If your supply trips current limit at startup, you may need a beefier source or a slow-start power switch upstream.

Thermals at 2 A

Rated 2 A continuous is achievable but the small QFN package gets warm. Above ~ 1.5 A continuous, consider airflow, a heatsink on the chip top, or thermal vias to a copper pour. Thermal shutdown kicks in around 150 °C junction temperature — well above what's safe for nearby components.

Reverse current

The TPS63070 can sink current from VOUT back to VIN under certain transients (this is normal and is what enables clean PWM operation at light loads). If your load includes batteries or other voltage sources on the output side, add a Schottky diode in series to prevent unwanted reverse flow.

Noise-sensitive loads

The 2.4 MHz switching ripple is small (typically tens of mV) but nonzero. For ADCs measuring low-level signals, RF receivers, or precision analog circuits, add a small LC filter or a downstream LDO (e.g. AP2112-3.3) to clean up the rail. The LDO's dropout is fine since it's regulating 3.3 V $\rightarrow$ 3.3 V (use a 3.6 V output board version, or use a different output voltage and step down).

9. Troubleshooting

No output / output reads 0 V: Check VIN polarity (most likely issue) — reverse polarity may have damaged the chip. Verify VIN is at least 2 V. Confirm all four pads are soldered cleanly.

Output reads low (e.g. 2.8 V) under load: Source can't deliver enough current — VIN is sagging. Check supply rating and use thicker / shorter input wires. A weak USB cable or coin cell will exhibit this.

Module gets hot quickly: Likely drawing close to 2 A, or you have a near-short on the output. Measure output current; if below 500 mA but still hot, check for solder bridges around the chip.

Audible whine from the inductor: Common in PFM mode at very light loads — the switching frequency drops into the audible range. Adding a small constant load (a few mA) usually shifts it back to PWM and silences it.

Connected MCU resets / browns out under load: Switching transients during heavy current pulses are dragging the rail. Add a 22 μ F ceramic right at the MCU's VCC pin, and check that GND wire is short and thick.

Output overshoots at startup: Normal during soft-start with a light load. If excessive, add output capacitance. If your load is voltage-sensitive (e.g. 3.3 V CMOS that doesn't tolerate 3.6 V), enable it only after the rail settles.

Module fails after one use with a 12 V battery: Was the battery actually < 16 V? Some 12 V SLA batteries float at 13.8 V (fine), but a freshly-charged 14.4 V LiFePO4 12-V pack can briefly exceed 14.6 V (still fine). 24 V supplies are *not* safe — they will destroy the chip.

Output is exactly equal to input below 3.3 V: The chip should still boost — but if the load is too heavy for the input source, the chip may be in current limit. Try a lighter load to confirm normal operation.

10. References & Further Reading

- **Texas Instruments — TPS63070 product page**

<https://www.ti.com/product/TPS63070>

- **Texas Instruments — TPS63070 datasheet (PDF)**

<https://www.ti.com/lit/ds/symlink/tps63070.pdf>

- **TI WEBENCH Power Designer (interactive design tool)**

<https://webench.ti.com/power-designer/>

- **TPS63070EVM-693 evaluation module user guide**

<https://www.ti.com/tool/TPS63070EVM-693>

- **Power Electronics News — 3 V-15 V to 5 V buck-boost using TPS63070**

<https://www.powerelectronicsnews.com/3v-15v-to-5v-1a-buck-boost-converter-using-tps63070/>

- **TI E2E Forum (community Q&A; for TPS63070 design issues)**

<https://e2e.ti.com/support/power-management-group/power-management>

Reference compiled from the official Texas Instruments TPS63070 datasheet and community sources. The Amazon listing's "XL63070" name is marketing; the actual silicon is TI's TPS63070. Always consult the current TI datasheet for authoritative specifications.